Harper Rodriguez

Senior Mechanical Engineer

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Natick, MA
LinkedIn

Senior Mechanical Engineer with 10+ years of experience in product design, FEA, and DFM using tools like SolidWorks, AutoCAD, ANSYS, and GD&T. Known for delivering efficient, manufacturable solutions across cross-functional teams.

WORK EXPERIENCE

MathWorks

Senior Mechanical Engineer

Natick, MA | 2021 - current

- Cultivated relationships with 5 key suppliers using Autopilot, cutting material lead times by 14%
- Identified and mitigated potential design risks using RiskOptics, preventing 12 post-launch issues across two product lines
- Led the development of three new products from concept to production, **generating $2.37 million in annual revenue**
- Reduced defect resolution time from 14 days to 6 days by resolving system integration issues

General Electric

Lead Mechanical Engineer

Boston, MA | 2018 - 2021

- Oversaw the training and development of seven engineers, with 97% of them advancing to higher roles within the company
- Implemented lean manufacturing principles, **reducing production time by 6.5 hours per unit** across two major product lines
- Collaborated with cross-functional teams on Notion to improve design integration, eliminating 9 recurring design-related delays per project cycle
- Improved Intellect QMS quality control processes, reducing product defects by 42 per 1,000 units

Raytheon Technologies

Principal Mechanical Engineer

Waltham, MA | 2015 - 2018

- Engineered a robotic assembly line for consumer electronics, **reducing assembly errors by 29%**
- Led an AutoCAD optimization project that cut production cycle time by 7.8 hours per unit, enhancing workflow efficiency
- Guided a team of 11 engineers in developing a next-gen HVAC system that lowered annual energy use by 1,270 kWh per unit installed
- Implemented Fusion 360 simulation, cutting prototype iterations from 9 to 5 and reducing validation time by 3 weeks

SKILLS

Software & Data Tools

- Python and Oracle

Systems & Automation

- SCADA / HMI Systems

Quality & Process Control

- Six Sigma (Green/Black Belt)

Manufacturing & Production Engineering

- Design for Assembly (DFA)

Production Processes

- Concept-to-Prototype Acceleration

EDUCATION

Bachelor of Science Mechanical Engineering

Massachusetts Institute of Technology

2011 - 2015

Cambridge, MA

Awards

- ASME, 2019

Professional Course

- MIT xPRO, 2023

CERTIFICATIONS/ LICENSES

- Professional Engineer (PE)